

## **Load Balancing in AWS -II**

### ❖ **Target Group**

- A **Target Group** in AWS is a logical grouping of targets (like EC2 instances, IP addresses, or Lambda functions) that a **Load Balancer** uses to route incoming traffic.
- When a Load Balancer receives a request, it uses listeners and routing rules to forward the request to a target group, which in turn directs the traffic to one or more registered target.

### ✓ **Supported Target Types:**

- **Instance** – EC2 instances (registered using instance ID)
- **IP** – Private IP addresses within or outside the VPC
- **Lambda** – AWS Lambda functions (for ALB only)
- **Application Container** – ECS tasks using the awsvpc network mode

### ✓ **Key Properties of a Target Group**

- **Target Type** :- Defines how targets are registered.
- **Protocol** :- Communication protocol between Load Balancer and targets.
- **Port** :- Port on which the target receives traffic.
- **Health Check Configuration** :- Used to monitor target health.

### ❖ **Listener**

- Listener is a process running on a Load Balancer that continuously checks for incoming connection requests on a specified **protocol** and **port**. Based on rules,
- The listener determines how to forward these requests to appropriate target group.

### ➤ **Key Components of a Listener**

#### 1. **Listener Protocol & Port**

- Defines how incoming connections are received.

#### 2. **Default Action (Forward Rule)**

- Determines what happens to the incoming request.
- Most common: Forward to a Target Group.

#### 3. **Rules**

- Allow conditional routing based on path, host, header, etc.
- Used only with **ALB**.

#### 4. **Priority (in rules)**

- Determines the order in which listener rules are evaluated.
- Lower number = higher priority.

#### 5. **Listener Rules (for ALB):**

##### ➤ *Listeners can evaluate incoming requests using rules and conditions such as:*

- **Host-based routing** – Route traffic based on the domain (**e.g.**, api.example.com)
- **Path-based routing** – Route based on the URL path (**e.g.**, /images, /login)
- **Header** or **query** string matching
- **Source IP** address matching

## ▪ Benefits of Using Listeners (AWS Load Balancer)

1. **Traffic Routing** :- Listeners direct incoming traffic to the correct target group based on protocol and port.
2. **Supports Multiple Protocols**:- Can handle HTTP, HTTPS, TCP, or UDP traffic as per the Load Balancer type.
3. **Flexible Request Routing (ALB)** :- Enables host-based or path-based routing (e.g., /api/\*, blog.example.com).
4. **SSL Termination (HTTPS Listeners)** :- Offloads SSL encryption from backend servers, reducing their load.
5. **Authentication Support (ALB)** :- Supports user authentication via Amazon Cognito or OIDC providers.
6. **Custom Actions** :- Can redirect requests or send fixed responses without reaching backend servers.
7. **Prioritized Rules** :- Allows fine-grained control using rule priority and multiple conditions.

### ➤ Supported Actions:

#### ✓ **Action Type Description**

- **Forward** :- Sends request to a specified target group
- **Redirect** :- Sends an HTTP 301/302 redirect to a new
- **URL Fixed Response** :- Returns a static HTTP response (e.g., 403 or 200 message)

## ❖ Health Checks

- A Health Check is a process used by AWS Load Balancers to monitor the status of targets (such as EC2 instances, containers, or Lambda functions) in a target group.
- Health checks help ensure that only **healthy** and **responsive targets receive** traffic . If a target fails a health check, it is **automatically marked as unhealthy**, and traffic is no longer routed to it until it recovers.

## ❖ Availability Zone Support

Availability Zone (AZ) Support refers to the ability of AWS Load Balancers to **distribute traffic across** multiple **Availability Zones** within a region, ensuring **fault tolerance** and **high availability** of applications.

### ❖ Availability Zone (AZ):

- An AZ is an **isolated** data center within a **region** that has **independent power, networking, and cooling**.
- Multiple AZs provide **redundancy** and failure isolation.
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#### 1. **How Load Balancers Use AZs:**

##### ▪ **Multi-AZ Load Balancing:**

- Multi-AZ Load Balancing means that the Load Balancer distributes traffic across targets (EC2, containers, etc.) that are located in multiple Availability Zones (AZs).
- If one AZ becomes unavailable, traffic is automatically routed to healthy instances in the other AZs.

##### ▪ **Cross-Zone Load Balancing:**

*Ensures traffic is distributed evenly across all registered targets, regardless of the AZ in which they reside.*

##### ▪ **Can be enabled or disabled:**

- Enabled (default for ALB): Load Balancer distributes traffic evenly to all targets in all AZs.
- Disabled (default for NLB): Load Balancer distributes traffic only to targets in the same AZ as the request.

## ❖ Security Groups

- *Virtual firewalls that control inbound and outbound traffic at the instance level (for ALB).*

##### ▪ **Can be applied to:**

- Load Balancer itself
- EC2 targets behind the Load Balancer

Define rules for allowed IP ranges, ports, and protocols.

## ❖ AWS WAF (Web Application Firewall)

*Protects applications from common web exploits such as:*

- SQL injection
- Cross-site scripting (XSS)
- Bot traffic
- Works with Application Load Balancer only.
- Allows setting rules to allow, block, or count specific request patterns

### ***[General Interview Questions]***

#### **1. What are target groups in ALB?**

- A collection of **EC2 instances**, **IPs**, or **Lambda** functions where traffic is routed. Each target group is linked to specific routing rules.

#### **2. What is a listener rule?**

- A condition defined for a listener that determines **how requests are routed** to target groups (**e.g., host-based or path-based**).

#### **3. What happens if no rules match a request in ALB?**

- The request is forwarded to the **default target group configured** for the listener.

#### **4. What is the role of health checks in ALB?**

- To **monitor target instance** health. If **a target fails health checks**, ALB stops routing traffic to it.

#### **5. What is the difference between internal and internet-facing ALB?**

- **Internet-facing** ALBs route **traffic** from the *internet*
- **Internal** ALBs route **traffic** only within *the VPC (private)*

#### **6. Can multiple target groups share the same EC2 instance?**

- **Yes**, if the instance is **registered** in **multiple target** groups or listens on different ports.

#### **7. How can you secure an ALB endpoint?**

- Use HTTPS listener with SSL certificate (via ACM)
- Use Security Groups to restrict access
- Use WAF for protection from web threats

#### **8. How would you handle blue/green deployment with ALB?**

- Create two target groups, **one** for **“blue”** (current version) and **one** for **“green”** (new version).
- Switch traffic gradually or instantly using listener rules.